

UNIVERSITY OF GHANA



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DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE ACTU 302: INTRODUCTION TO ACTUARIAL COMPUTING(3 CREDITS) EXERCISE I

Study the functions `ExpQQ` and `ParetoQQ` in the R package `ReIns`.

- i. What does the functions, `ExpQQ` and `ParetoQQ`, outputs show? Explain.
- ii. Generate a number of observations from the normal distribution with mean 20 and variance 4. Use the functions in (i) to plot these data. Repeat this exercise but in this case make sure there is no title in your plots.
- iii. Look for the cumulative distribution functions of the exponential and Pareto distributions. Invert these functions to obtain the quantile functions for each distribution.
- iv. Write your own functions based on (iii) to obtain the plots in (ii) and compare them.
- v. Simulate 100 observations from the exponential distribution with mean 4. Use your functions in (iv) to obtain plots of this data. Comment of the fit of these distributions to the data.
- vi. The Society of Actuaries (SOA) Group Medical Insurance data studied in Minkah et al. (2021) and can be found at <https://lstat.kuleuven.be/Wiley/Data/soa.txt>. Read the data into R and download the paper. Reproduce the plots concerning the claims data on the left panel Figure 10 Minkah et al. (2021, pg. 16) and left panel of Figure 11 Minkah et al. (2021, pg. 17). Comment on the fit of these distributions to the data

References

Minkah, R., de Wet, T., and Ghosh, A. (2021). Robust estimation of Pareto-type tail index through an exponential regression model. *Communications in Statistics - Theory and Methods*, pages 1–21.